

END SEM: MTH201

Name: [REDACTED]

Reg No.: [REDACTED]

Time: 3 h Marks: 54

Q 1. Check if the following limit exists. If it exists, then calculate the limit.

(i) $\lim_{x \rightarrow -1} \frac{|x+1|}{(x+1)x}$ [3]

(ii) $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right)$ [2]

Q 2. Consider the function $f(x) = \begin{cases} 3x - 5 & \text{if } x \neq 1 \\ 2, & \text{if } x = 1 \end{cases}$.

(i) Determine if the function $f(x)$ is continuous at $x = 1$. [3]

(ii) Find the largest intervals where f is strictly increasing. [2]

Q 3. Let $f(x) = x$. For the interval $[0, 1]$, consider the partition $P_n = \{0, 1/n, 2/n, 3/n, \dots, n/n\}$.

(i) Calculate the upper sum $U(P_n, f)$. [2.5]

(ii) Calculate the upper integral which is the limit of $U(P_n, f)$ as $n \rightarrow \infty$. [2.5]

Q 4. Calculate $\int_{-\pi/2}^{\pi/2} (|x| + |\sin(x)|) dx$. [2.5]

Q 5. Show that, for any curve defined by a vector-valued function $r(t)$, the unit tangent vector $T(t)$ is always orthogonal to its derivative $T'(t)$. [2.5]

Q 6. Consider the curve C defined by $\mathbf{r}(t) = (\cos t + t \sin t)\mathbf{i} + (\sin t - t \cos t)\mathbf{j}$, for $t > 0$. Calculate the unit tangent vector $T(t)$, the unit normal vector $n(t)$, and the curvature $\kappa(t)$ at points on C . [6]

Q 7. Consider the function

$$f(x, y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Show that this is not continuous at $(0, 0)$. [4]

Q 8. Consider the function $g(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0); \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$

(i) Calculate the partial derivatives at $(0, 0)$. [2]

(ii) Show that $g(x, y)$ is not differentiable at $(0, 0)$. [2]

Q 9. Consider the parallelogram with the vertices $(2, 0), (5, 3), (6, 7), (3, 4)$. Calculate the integral $\iint_R (6x - 3y) dx dy$, using the substitution $x = (v - u)/3, y = (4v - u)/3$ in R . [5]

Q 10. Let C be the semi-circle given by $x^2 + y^2 = 4, y \geq 0$, oriented from the left to the right.

(i) Calculate $\int_C ds$, where ds is an arc length element. [2]

(ii) Let $\mathbf{F}$ be the gradient of the scalar field $f(x, y) = \log(x^2)$. Calculate the integral $\int_C \mathbf{F} \cdot d\mathbf{r}$. [2]

Q 11. Find the absolute minimum and maximum for the function $f(x, y) = x + 2y^2 + 1$ on the boundary of the unit disk $C = \{x^2 + y^2 = 1\}$. Use Lagrange multipliers. [4]

Q 12. State True or False. Proof not needed.

(i) The limit of a function always exists at a point. [1]

(ii) All continuous functions on a domain D are differentiable in D . [1]

(iii) All continuous functions on an open interval are integrable. [1]

(iv) Let $f(x, y)$ be a function such that the partial derivatives exist at a point (a, b) . Then f is continuous on (a, b) . [1]

(v) There exists a function $f(x)$ defined on the interval $[-1, 1]$ such that $|x| = \int_{-1}^x f(t) dt$. [1]

(vi) All differentiable functions on a bounded interval are integrable. [1]

(vii) If the derivative of $f(x, y)$ exists at a point (a, b) , then all the partial derivatives exist. [1]